Pollard Rho Across a Large Prime Conductor Gap: A Large-Sample Measurement and a Formal Invariance

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Abstract

Jao, Miller and Venkatesan proved that elliptic curves in one isogeny class over a prime field have equally hard discrete logarithms, up to polynomial factors, whenever the conductors of their endomorphism rings differ only at small primes. A gap at a *large* prime ℓ, bridged only by isogenies of degree divisible by ℓ, is the one case their expander argument does not reach, and whether the two sides can differ has been open since 2005. We measure the question for Pollard rho, the best known attack over prime fields, at the largest sample size to date: 24 audited sessions of a native harness, 87,000 verified solves, on CM curves of discriminant -19 and prime order, the crater ($\operatorname{End} E = \mathcal{O}_K$) against floors ($\operatorname{End} E = \mathbb{Z} + \ell\mathcal{O}_K$), under a protocol frozen before data with isomorphic-model negative controls and a $\sqrt{2}$ positive control. Where the gap is hard to cross (kernel field degree 2736 to $\ell - 1$; $\ell = 8209$, 131101 and $\approx 2^{20}$) the floor/crater ratio of charged group operations is 0.998 [0.985, 1.011] over 12,288 paired solves, 1.005 [0.960, 1.052] and 0.982 [0.940, 1.028], closing an earlier unexplained lead of 1.22. Where the gap is cheap to cross ($\ell \approx 2^{25}$ at 53 bits) a persistent 8 % deficit reproduced under fresh targets but vanished between re-coordinatised endpoints (1.040 [0.965, 1.119]), so it belongs to the coordinate models under the walk's representation hash, not to the endomorphism ring. We complement the measurement with a machine-checked theorem (Lean 4, no Mathlib, axioms `propext` and `Quot.sound`): transporting an r-adding walk along any group isomorphism preserves its trajectory and ρ-length and is a bijection of the space of partition and multiplier choices, so representation-hashing rho is conductor-blind in distribution and a fixed partition is the only channel left, which is what the measurement bounds. We state precisely what remains open: non-generic algorithms are bounded only by the cost of the connecting isogeny, and a proof of equal hardness is unavailable for any curve.

1 Introduction

Let E and E' be ordinary elliptic curves over a prime field $\mathbb{F}_q$ with the same number of points N. By Tate's theorem they are isogenous over $\mathbb{F}_q$, and any $\mathbb{F}_q$-isogeny $\varphi\colon E \to E'$ carries a discrete-logarithm instance $(P, Q = [k]P)$ on E to the instance $(\varphi P, \varphi Q)$ on E' with the same answer. If φ can be evaluated cheaply, the two discrete logarithm problems are equivalent. Jao, Miller and Venkatesan [5, 6] showed, assuming the Generalised Riemann Hypothesis, that the graph whose vertices are the curves of an isogeny class with a fixed endomorphism ring and whose edges are isogenies of small prime degree is an expander, so a short random walk of cheap isogenies connects any two such curves: within one level of the isogeny volcano [7, 4, 12] the ECDLP has a single difficulty up to polynomial factors. Their result extends across levels whose endomorphism conductors differ at small primes, because a vertical step of small degree is cheap. It stops at a *large prime conductor gap*: if $\operatorname{End} E = \mathcal{O}_K$ and $\operatorname{End} E' = \mathbb{Z} + \ell\mathcal{O}_K$ for a large prime ℓ, every isogeny between E and E' has degree divisible by ℓ, and no polynomial-time (in $\log \ell$) way to evaluate one is known for a non-rational kernel. The authors note the case explicitly and leave open whether the two sides can differ in difficulty.

The question has two parts. The *structural* part asks whether any algorithm can exploit the endomorphism ring; no proof either way is available, since it would need lower bounds on the ECDLP that exist for no curve. The *empirical* part asks whether the attacks we have, run on the two sides of such a gap, cost the same. Over prime fields the best attack is Pollard rho [10] with the negation map [2] and parallel collision search [13]; its cost depends on the group order alone in the generic model [9, 11], but the implementations people run are not generic: the partition that chooses the next step hashes the *representation* of the point, which an isomorphism of curves changes. Whether that hash lets the conductor leak into the cost is therefore a question about the implementation, and it can only be answered by measurement.

This paper reports such a measurement at a sample size large enough to settle it to the one-percent

level, with a design that separates the conductor from the representation, and complements it with a machine-checked statement of exactly what can be proved about representation-hashing walks. Every number cites a committed artefact of the companion research repository (prefix `crypto:`); the study directory is `crypto:research/conductor_gap_rho_large_sample_20261018/`.

Contributions.

1. A frozen-protocol measurement of Pollard rho on 87,000 verified discrete logarithms across prime conductor gaps at $\ell \in \{8209,\ 131101,\ 1{,}138{,}019,\ 1{,}211{,}653,\ 1{,}609{,}903\}$ in the regime where the gap cannot be crossed for less than the cost of the DLP, and at $\ell \approx 2^{25}$ where it can. The floor/crater ratio of charged group operations is $0.998\,[0.985, 1.011]$, $1.005\,[0.960, 1.052]$ and $0.982\,[0.940, 1.028]$ in the three hard-gap families. An earlier lead of $1.22\,[1.005, 1.46]$ at $\ell = 8209$ is closed.
2. A control design (isomorphic twins of both endpoints, a $\sqrt{2}$ positive control, shared targets for pairing) that attributes the one persistent shift, an 8 % deficit of the floors at $\ell \approx 2^{25}$ reproduced under fresh targets, to the specific coordinate models used and not to the endomorphism ring: between re-coordinatised endpoints the same gap measures $1.040\,[0.965, 1.119]$.
3. A theorem checked in Lean 4 without Mathlib: along any isomorphism of groups, an r-adding walk whose partition hashes the representation has a transported partner with the same trajectory and ρ-length, and the transport is a bijection of the walk space. Over a uniformly random partition, rho's cost distribution is therefore identical on the two sides of any gap; a fixed partition is the only channel the conductor can use, and the measurement bounds it.

2 Setting

2.1 Conductor gaps

Let $K = \mathbb{Q}(\sqrt{D_K})$ be an imaginary quadratic field of fundamental discriminant D_K and let $E/\mathbb{F}_q$ be ordinary with trace t. Then $t^2 - 4q = f^2 D_K$ for an integer f, and $\operatorname{End} E = \mathbb{Z} + c\,\mathcal{O}_K$ is an order of conductor $c \mid f$. Curves of trace t form the vertices of the ℓ-isogeny volcano for each prime ℓ; the conductor's ℓ-adic valuation is the depth. We fix $D_K = -19$, so $h(D_K) = 1$, the unit group is ± 1 on every curve (no automorphism confound) and the crater consists of the single j-invariant $j_K = -884736$. We choose $f = \ell$ prime, so that every curve of trace t has conductor 1 (the crater) or ℓ (a floor), and every isogeny between the two levels has degree divisible by ℓ.

The construction is the one of the companion repository's earlier studies: $q = (t^2 + 19\ell^2)/4$ and $N = q + 1 - t$ both prime, the crater the twist of the j_K-model with N points, the floors any other curve of order N, certified by $[N]P = O$ for an affine point (which suffices because N is prime and exceeds the Hasse width). No isogeny is computed.

2.2 How hard is the gap to cross?

Let $\lambda = t/2 \bmod \ell$. Since $t^2 - 4q \equiv 0 \pmod{\ell^2}$, Frobenius acts on $E[\ell]$ of the crater as the scalar λ [8], every one of the $\ell + 1$ descending ℓ-isogenies is $\mathbb{F}_q$-rational as a subgroup, and the points of each kernel are defined over $\mathbb{F}_{q^k}$ with k the order of λ in $\mathbb{F}_\ell^\times$. We call k the *kernel field degree*. When k is small, a kernel point can be found over a small extension and the isogeny computed by Vélu's formulas [14] in $O(\ell)$ or by $\sqrt{\text{élu}}$ [1] in $\tilde{O}(\sqrt{\ell})$ operations: the gap is *easy*. When k is large the kernel polynomial must be found by factoring the ℓ-division polynomial or by the modular polynomial; the companion repository measured the scaling of its own implementation as $\ell^{2.605}$ (`crypto:benchmarks/20261011-isogeny-vertical-margin-historical`). We call such a gap *hard* and quantify it against the rho cost $\sqrt{\pi N/4}$: at $\ell = 8209$ with q of 29 bits the crossing costs about 2^{34} operations against a rho solve of $2^{14.5}$; at $\ell \approx 2^{20}$ with q of 45 bits, 2^{53} against $2^{22.5}$. In the hard regime the transfer argument of [5] gives no information, which is why it is the regime that matters. The easy regime ($t = \ell - 2$, so that $\lambda = -1$ and the quadratic twist has rational ℓ-torsion, $k = 2$) is included because it is where the companion repository's largest ℓ and its one unexplained rho value live.

2.3 The measured algorithm and its unit

All measurements use the repository's native harness `ecbench` and its method `rho.negation`: a 32-adding walk on negation classes with distinguished points, parallel collision search of van Oorschot and

tier	ℓ	q bits	k	gap	curves	targets	solves
S1 (4 shards)	8209	29	2736	hard	crater, 3 floors, 2 twins	4×1024	73,728
S2	131101	37	65550	hard	crater, 1 floor, 2 twins	1024	12,288
M1, M2, M3	1,211,653, 1,138,019, 1,609,903	45	$\ell-1$; $\ell-1$; $(\ell-1)/2$	hard	crater, 3 floors each	128	3,072
L	33,555,419 to 35,282,963	53	2	easy	24 pairs, 8 crater twins	16	896
L replication	same	53	2	easy	same, fresh targets	16	896
L twin round	same	53	2	easy	24 pairs, twins of both ends	16	1,536

Table 1: Families. Tier S curves are the Sage-verified pairs of `crypto:research/conductor_gap_cm_pairs_20261010`; tier M parameters are new and are built and certified inside the session by the harness's native constructor; tier L pairs are entries 0–23 of `crypto:research/conductor_gap_cm_orbits_20261015/inputs/pairs.json`. "Twins" are isomorphic models (au^4, bu^6) with generator (u^2x, u^3y), $u = 3$ (tiers S, L) or $u = 5$ (twin round). Solve counts include every arm.

Wiener, and the cycle handling of Bernstein, Lange and Schwabe [2]. The next step is chosen by a hash of the canonical representation of the current point. The unit is S, the number of charged group additions and doublings divided by $\sqrt{r}$ with $r = N$ the prime order; the generic expectation with the negation map is $\sqrt{\pi/4} \approx 0.886$. The single unpriced counter, canonicalisations, is the same on every arm, so the repository's pricing band does not change any ranking. Wall time was never read: the host was shared with other jobs and every run was recorded at the lowest isolation level. Every run is deterministic given its seeds; every session was re-audited by the harness (`ecbench verify`), with every record's answer checked against the planted logarithm and runs re-executed with identical counts.

3 Design

The protocol (`crypto:research/conductor_gap_rho_large_sample_20261018/PROTOCOL.md`, frozen with SHA-256 digests before any session ran) fixes the null hypothesis H_0: for every (q, N) the per-target distribution of S is the same on the crater and on every floor. It fixes the families of Table 1, the arms, the seeds, the unit, the analysis and the decision rule, and lists what may not change after data are seen. Two pre-data amendments (a harness mode that produced no sessions, and a sharding of the largest session without changing its total) and two pre-registered post-hoc follow-ups are recorded in `AMENDMENTS.md`.

Controls. Three controls were planned. The *negative control* is the ratio between a curve and its isomorphic twin: same endomorphism ring, same group, different point representation. Since the walk's partition hashes the representation, this ratio is the null distribution of "different coordinates, everything else equal"; a floor/crater ratio inside the twin band of its tier cannot be attributed to the conductor. The *positive control* is `rho.plain` (no negation map) against `rho.negation` on the same workloads, with expected ratio $\sqrt{2}$; if it is not seen the detector is broken. An *A/A arm* repeats `rho.negation`. The harness shares the algorithm seed per workload across arms, so the A/A arm reproduced the reference exactly on every target; it is a determinism check, and the between-draw noise floor is supplied by the twins instead.

Pairing and sample size. Targets are *planted and shared*: the same scalars on every curve of a session, so every comparison is paired per target. With the standard deviation of S for rho at about 0.52 of its mean, n targets per curve give a standard error of about $0.52\sqrt{2/n}$ on a ratio: 1.2 % at $n = 4096$ (S1 pooled), 2.3 % at 1024, 6.5 % at 128, and 3.7 % for tier L pooled over 24 pairs. The earlier lead at $\ell = 8209$ (1.22 from four targets) is resolved at the one-percent level. Seeds are fresh: the measurement seed is 20261018 where every earlier study used 1 or 20261012, and each session has its own target seed.

Analysis and decision rule. Per session and arm: mean S per curve, the ratio floor/crater and twin/original with a 95 % bootstrap interval over targets (4000 draws), and the paired per-target log-ratio with a normal 95 % interval. Pooled ratios per tier concatenate the per-target values of all floors and all craters of the tier and are never pooled across tiers. H_0 is rejected for a tier only if (a) a floor/crater interval excludes 1 after a Bonferroni correction over the floors of the tier, (b) the twin intervals of the tier contain 1, and (c) the positive control shows $\sqrt{2}$.

family	pairs	solves / side	ratio of means	95 %	paired geo-mean [95 %]
S = S1 + S2	13	13,312	**0.998**	[0.986, 1.010]	0.999 [0.985, 1.014]
S1 ($\ell = 8209$)	12	12,288	0.998	[0.985, 1.011]	1.000 [0.985, 1.015]
S2 ($\ell = 131101$)	1	1,024	1.005	[0.960, 1.052]	0.992 [0.939, 1.048]
S twins / originals	10	10,240	1.004	[0.990, 1.019]	1.004 [0.987, 1.021]
M = M1–M3 ($\ell \approx 2^{20}$)	9	1,152	**0.982**	[0.940, 1.028]	0.994 [0.941, 1.050]
M1 / M2 / M3	3 each	384	1.050 / 0.967 / 0.933		

Table 2: Pooled floor/crater for the hard-gap tiers, arm `rho.negation`. "Ratio of means" is $\overline{S}_{\text{floor}}/\overline{S}_{\text{crater}}$ with a bootstrap interval over targets; the paired column is the geometric mean of per-target ratios on shared targets. Source: `results/summary.md`, pooled table.

round	design	floor / crater	crater-twin / crater	floor-twin / floor	floor-twin / crater-twin
L frozen	24 pairs, 8 crater twins ($u = 3$)	0.917 [0.850, 0.988]	1.039 [0.919, 1.176]		
replication	same curves, fresh targets	0.920 [0.856, 0.993]	0.891 [0.791, 1.005]		
L + replication	48 pair-samples	0.919 [0.871, 0.970]			
twin round	twins of both ends ($u = 5$)	0.945 [0.878, 1.021]	0.967 [0.898, 1.042]	1.064 [0.987, 1.146]	**1.040** [0.965, 1.119]

Table 3: Tier L and its two pre-registered follow-ups, arm `rho.negation`, ratio of means with bootstrap 95 % intervals, 384 solves per side in each pooled cell (768 for L + replication). Source: `results/summary.md`, pooled rows `l`, `lr`, `l+lr`, `t` and their twin keys.

4 Results

All numbers are from `crypto:research/conductor_gap_rho_large_sample_20261018/results/summary.json` and the per-session run reports in `reports/run-*/`; every record of every session is `verified`. The 24 audit receipts in `audits/` are all `ok` with zero problems: the tier S and M sessions with every run re-executed (`--replay-all`; for instance 18,432 of 18,432 identical in each S1 shard, 1,024 of 1,024 in each M session), the 53-bit sessions with 24 identical replays each.

4.1 Hard gaps: tiers S and M

Table 2 and Figure 1 give the pooled ratios. Of the fifteen floor/crater comparisons in tier S ($n = 1024$ each) none excludes 1; they range from 0.968 to 1.017. Of the nine in tier M ($n = 128$) none excludes 1. The twin controls of tier S pool to $1.004\,[0.990, 1.019]$, with crater-twin/crater at $1.008\,[0.988, 1.028]$ and floor-twin/floor at $1.001\,[0.981, 1.022]$: re-coordinatising a curve moves rho's mean cost by at most about two percent at this sample size, and every floor/crater ratio is inside that band.

The positive control was seen in every session: `rho.plain` over `rho.negation` on the crater is $1.360\,[1.299, 1.423]$, $1.372\,[1.314, 1.432]$, $1.424\,[1.363, 1.487]$ and $1.393\,[1.334, 1.452]$ in the four $\ell = 8209$ shards and $1.446\,[1.383, 1.510]$ at $\ell = 131101$ (Figure 2). Every interval excludes 1 and brackets $\sqrt{2}$; the detector sees a 41 % effect without difficulty, so the absence of a floor/crater effect is not an absence of power.

4.2 The easy gap: tier L and its follow-ups

The frozen tier L (24 pairs at $\ell \approx 2^{25}$, 16 targets per curve) pooled to $0.917\,[0.850, 0.988]$: the floors used fewer operations than the craters, in the opposite direction to the companion repository's earlier post-hoc value on the same pairs ($1.17\,[1.00, 1.38]$, one target each). The frozen rule did not reject H_0: no single pair survived the Bonferroni correction (two of 24 excluded 1 on their own), and one of the eight twin controls in the tier excluded 1 in the other direction ($1.59\,[1.20, 2.18]$). Two follow-ups were then pre-registered and run on the same pairs with fresh target seeds (Table 3).

The replication reproduced the frozen ratio almost exactly (0.920 against 0.917), so the deficit is not a single unlucky draw; over 48 pair-samples the floors cost 8 % [3 %, 13 %] fewer operations than the craters. But the replication's twin control moved by the same amount with no change of endomorphism ring ($0.891\,[0.791, 1.005]$), and the pairs whose intervals excluded 1 were driven by high-S crater draws that depressed the twin ratio of the same pair just as much. The twin round was designed to separate the two readings. Its three pre-registered comparisons all contain 1, and the decisive one, floor-twin against crater-twin, is $1.040\,[0.965, 1.119]$: the same conductor gap between two freshly re-coordinatised models shows no deficit at all, while the comparisons involving the original crater models (0.945, 0.967) keep a few percent of it. The shift attaches to the particular coordinate models of the pair file (all 24 craters

are the same j_K model family, constructed the same way), not to $\mathrm{End}\,E$. What remains is a statement about the harness's fixed partition hash: on specific models it can move rho's mean cost by about five percent at $n = 384$, which is exactly the residual that Section 5 says a fixed hash can carry and that a uniformly random hash cannot. The earlier post-hoc 1.17 on these pairs is closed the same way.

5 A formal invariance for representation-hashing walks

The generic-group argument [9, 11] says that an algorithm whose control flow depends only on equality answers has the same transcript on isomorphic groups; the companion repository had already machine-checked this (`crypto:research/conductor_gap_cm_pairs_20261010/lean/GenericInvariance.lean`), and an integrity audit had pointed out that it does not cover the rho and kangaroo the harness runs, because they hash the point representation. We close that gap in the only form in which it is true: not for one fixed hash, but for the space of hashes.

Definition 1. *Let G be a type with a binary operation $\cdot$ (no axioms are used). An r-adding walk on G is a pair $w = (h, m)$ with $h\colon G \to \{0, \dots, r-1\}$ (the partition of the representation) and $m\colon \{0, \dots, r-1\} \to G$ (the multipliers); its step is $x \mapsto x \cdot m(h(x))$, its trajectory from x of length n is $[x, \mathrm{step}\,x, \dots, \mathrm{step}^n x]$, and its ρ-length with budget B, $\rho_w(x, B)$, is the least $n \le B$ whose trajectory contains a repeated element, or $B + 1$ if none does. Given a bijection $\varphi\colon G \to H$ with $\varphi(a \cdot b) = \varphi a \cdot \varphi b$, the* transport *of w is the walk $\varphi_* w = (h \circ \varphi^{-1},\ \varphi \circ m)$ on H.*

Theorem 1 (`rhoCost_transport`, `costPreservingBijection`)**.** *For every walk w on G, start $x \in G$ and budget B, the trajectory of $\varphi_* w$ from φx is φ applied pointwise to the trajectory of w from x, and $\rho_{\varphi_* w}(\varphi x, B) = \rho_w(x, B)$. The map $w \mapsto \varphi_* w$ is a bijection from the walks on G to the walks on H, with inverse $(\varphi^{-1})_*$.*

Proof. One step commutes: $\varphi(x \cdot m(hx)) = \varphi x \cdot \varphi(m(hx)) = \varphi x \cdot (\varphi \circ m)((h \circ \varphi^{-1})(\varphi x))$. Induction gives the trajectory statement. A list has a repeated element if and only if its image under an injective map does, and φ is injective, so the first index with a repeat is the same; hence the ρ-lengths agree for every budget. The two transports compose to the identity on both sides because $\varphi^{-1} \circ \varphi = \mathrm{id}$ and $\varphi \circ \varphi^{-1} = \mathrm{id}$. The formal proof (`crypto:research/conductor_gap_rho_large_sample_20261018/lean/RepresentationHashInvariance.lean`, Lean 4.34.1, no dependencies) follows exactly these lines, with the collision test written as a decidable Boolean on lists so that no classical choice is needed; `#print axioms` reports `propext` and `Quot.sound` only (`lean/build.log`). □

Remark 1 (What the theorem says about the conductor). Two curves of the same prime order in one isogeny class have isomorphic groups of points, so Theorem 1 applies with any isomorphism φ between them. Because the transport is a bijection of the walk space, the push-forward of the uniform distribution on (h, m) is uniform: over a uniformly random partition and random multipliers, the distribution of rho's ρ-length is identical on the crater and on the floor. The endomorphism conductor, the j-invariant or any other invariant that is not an invariant of the abstract group can enter rho's cost only through the one fixed partition an implementation uses. That residual is what Section 4 measures: at most about two percent at $n = 1024$ on the twins, and about five percent on the specific models of tier L.

Remark 2 (What the theorem does not say). Nothing about curves, isogenies or endomorphism rings is formalised; the bridge "isogenous curves of equal prime order have isomorphic groups" is an elementary fact used outside the file. The theorem is silent about algorithms that are not walks of this shape, and about any fixed hash. Together with the repository's transfer bound (`TransferBound.lean`: any solver on one side yields a solver on the other at the cost of two isogeny evaluations), the formal picture is: generic algorithms are blind to the gap by theorem; representation-hashing walks are blind in distribution by theorem and blind in practice by measurement; any other algorithm can differ across the gap by at most the cost of the connecting isogeny.

6 Discussion

What is settled. For Pollard rho with the negation map, as implemented in a native, audited harness, the floor and the crater of a prime conductor gap are indistinguishable at the one-percent level for $\ell = 8209$ (29-bit q), at the five-percent level for $\ell = 131101$ (37-bit) and at the six-percent level for $\ell \approx 2^{20}$ (45-bit), all with a hard kernel (field degree 2736 to $\ell - 1$), under a fresh seed, with the positive

control confirming the detector and the twin controls bounding the representation effect at the same level. The companion repository's one open rho lead is closed, and so is its one post-hoc value at $\ell \approx 2^{25}$, by attribution to the coordinate models.

Why the result is not surprising, and why it had to be measured. In the generic model the answer is forced by the group order; what is not forced is the behaviour of a walk whose partition reads coordinates. The twin controls quantify that non-generic residual directly, and the tier L episode shows that it is real at the few-percent level for specific models: the same deficit appeared twice under independent targets, and only a design that re-coordinatised both endpoints could tell it apart from a conductor effect. Earlier leads in both directions on the same curves, with four and with one target per curve, were the same residual seen through too few samples.

What is not settled. Whether some non-generic algorithm exploits $\operatorname{End} E = \mathbb{Z} + \ell\mathcal{O}_K$ against $\mathcal{O}_K$. The transfer bound says any such advantage is at most the cost of one ℓ-isogeny evaluation, which for a non-rational kernel scales as $\ell^{2.6}$ in the repository's measurement; nobody has a cheaper way, and nobody has proved there is none (computing the endomorphism ring itself is subexponential [3], but evaluating a specific vertical isogeny of large prime degree is not known to be). A proof of equal hardness in the sense of the title question would need lower bounds on the ECDLP that exist for no curve. Index calculus is not measured here; the companion repository reports $0.980\,[0.866, 1.111]$ for it at $\ell \approx 2^{25}$. Everything here is over prime fields, $D_K = -19$, $q \leq 53$ bits and one walk.

Classification. Under the companion repository's rules this is a null result on a measured question, with one observation about the harness's partition hash; it is not an advance, an engineering gain, a relabelling or an accounting change, and it makes no security claim about any deployed curve.

7 Reproduction

`make_specs.py` writes the specifications; `run_sessions.sh` runs one harness session per specification; `audit.sh` produces the receipts; `analyze.py` is the fixed analysis; `plot.py`, `make_run_reports.py` and `make_report.py` lay out the recorded values. The 19 sessions of tiers M, L and the follow-ups are committed raw; the five tier-S sessions (420 MB) are committed as unmodified tarballs with checksums alongside their full-replay receipts. Two harness pitfalls met on the way are recorded with the amendments: an explicit CPU list made this build of the harness exit without a session, and per-execution process overhead of about 0.3 s dominates small-field sessions.

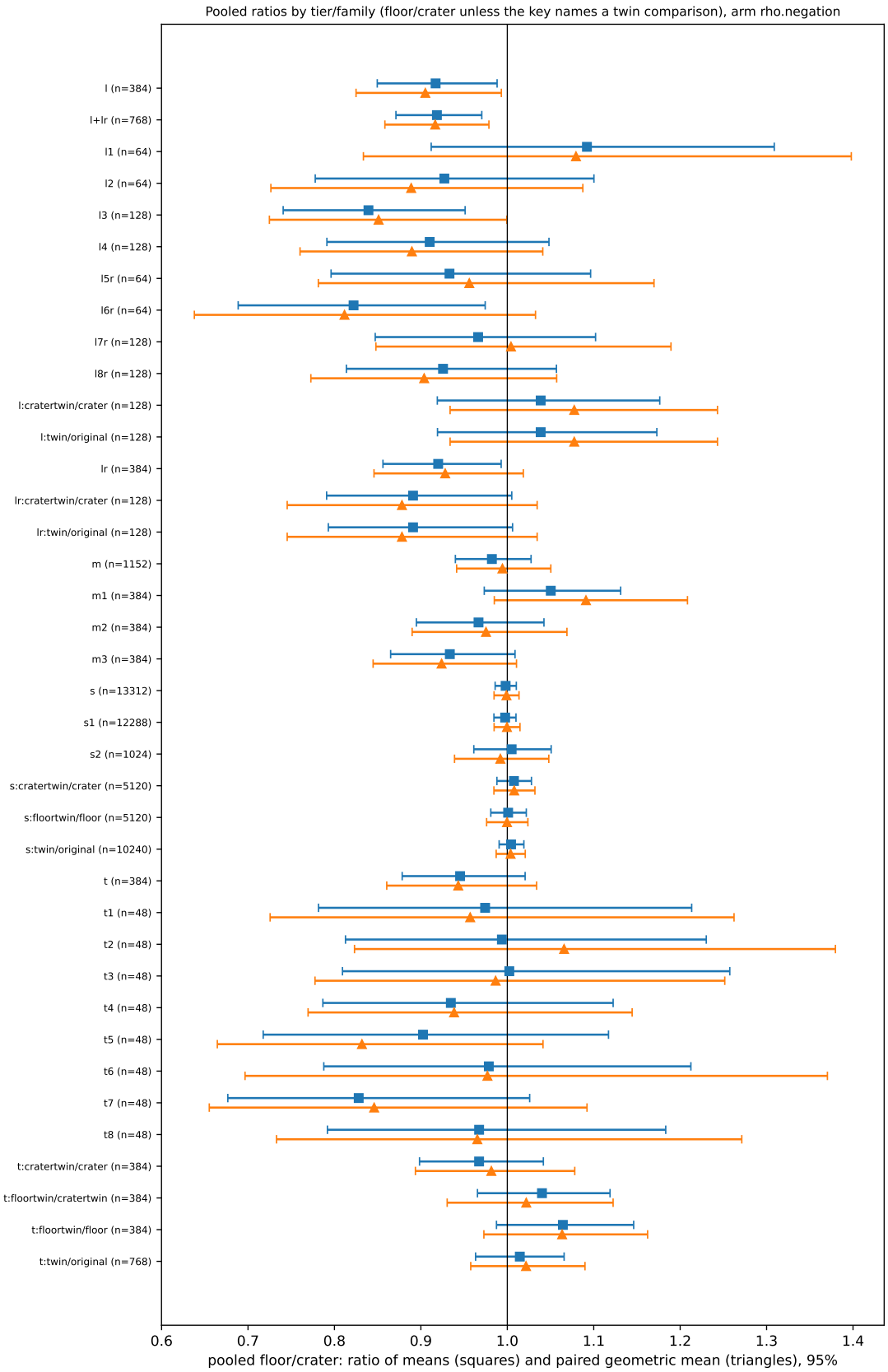


Figure 1: Pooled ratios by tier and family, arm `rho.negation`: ratio of means (squares, bootstrap 95 %) and paired geometric mean (triangles). Keys without a colon are floor/crater; keys with a colon are the named twin comparison. `s`: $\ell = 8209$ and 131101; `m`: $\ell \approx 2^{20}$; `l`, `lr`, `t`: the frozen, replication and twin rounds at $\ell \approx 2^{25}$. Source: `figures/pooled_ratios.svg`, plotted from `results/summary.json`.

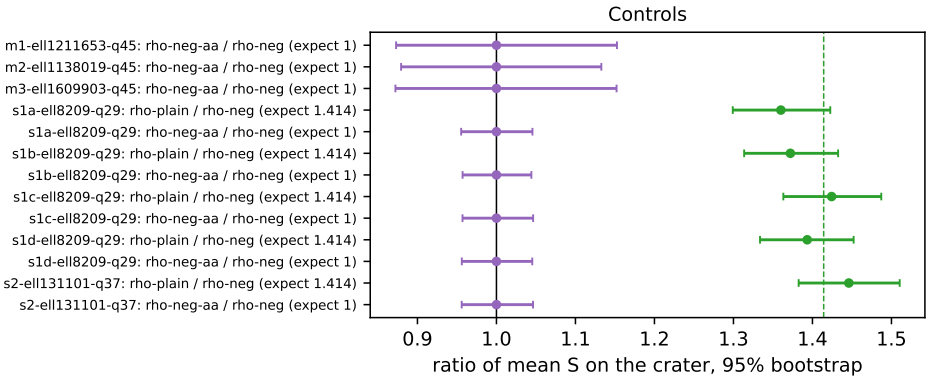


Figure 2: Controls on the crater of each tier S and M session: the positive control `rho.plain/rho.negation` (expected $\sqrt{2}$, dashed) and the A/A arm, which shares the algorithm seed with the reference and therefore reproduces it exactly. Source: `figures/controls.svg`.

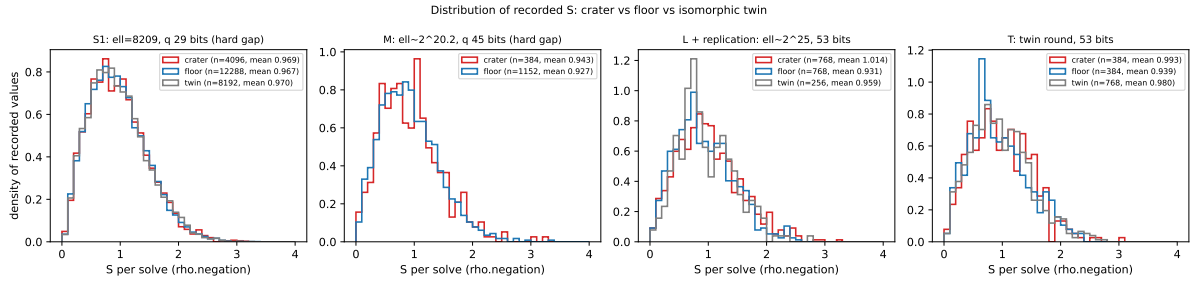


Figure 3: Distribution of recorded S per solve, crater against floor against isomorphic twin, in the $\ell = 8209$ shards, tier M, tier L with its replication, and the twin round. Source: `figures/s_distributions.svg`, plotted from the session records.

References

[1] Daniel J. Bernstein, Luca De Feo, Antonin Leroux, and Benjamin Smith. Faster computation of isogenies of large prime degree. *The Open Book Series (ANTS XIV)*, 4(1):39–55, 2020.

[2] Daniel J. Bernstein, Tanja Lange, and Peter Schwabe. On the correct use of the negation map in the Pollard rho method. In *Public Key Cryptography – PKC 2011*, volume 6571 of *Lecture Notes in Computer Science*, pages 128–146. Springer, 2011.

[3] Gaetan Bisson and Andrew V. Sutherland. Computing the endomorphism ring of an ordinary elliptic curve over a finite field. *Journal of Number Theory*, 131(5):815–831, 2011.

[4] Mireille Fouquet and François Morain. Isogeny volcanoes and the SEA algorithm. In *Algorithmic Number Theory – ANTS-V*, volume 2369 of *Lecture Notes in Computer Science*, pages 276–291. Springer, 2002.

[5] David Jao, Stephen D. Miller, and Ramarathnam Venkatesan. Do all elliptic curves of the same order have the same difficulty of discrete log? In *Advances in Cryptology – ASIACRYPT 2005*, volume 3788 of *Lecture Notes in Computer Science*, pages 21–40. Springer, 2005.

[6] David Jao, Stephen D. Miller, and Ramarathnam Venkatesan. Expander graphs based on GRH with an application to elliptic curve cryptography. *Journal of Number Theory*, 129(6):1491–1504, 2009.

[7] David Kohel. *Endomorphism rings of elliptic curves over finite fields*. PhD thesis, University of California, Berkeley, 1996.

[8] Hendrik W. Lenstra, Jr. Complex multiplication structure of elliptic curves. *Journal of Number Theory*, 56(2):227–241, 1996.

[9] Vassiliy I. Nechaev. Complexity of a determinate algorithm for the discrete logarithm. *Mathematical Notes*, 55(2):165–172, 1994.

[10] John M. Pollard. Monte Carlo methods for index computation (mod p). *Mathematics of Computation*, 32(143):918–924, 1978.

[11] Victor Shoup. Lower bounds for discrete logarithms and related problems. In *Advances in Cryptology – EUROCRYPT '97*, volume 1233 of *Lecture Notes in Computer Science*, pages 256–266. Springer, 1997.

[12] Andrew V. Sutherland. Isogeny volcanoes. *The Open Book Series*, 1(1):507–530, 2013.

[13] Paul C. van Oorschot and Michael J. Wiener. Parallel collision search with cryptanalytic applications. *Journal of Cryptology*, 12(1):1–28, 1999.

[14] Jacques Vélu. Isogénies entre courbes elliptiques. *Comptes Rendus de l'Académie des Sciences de Paris, Série A*, 273:238–241, 1971.